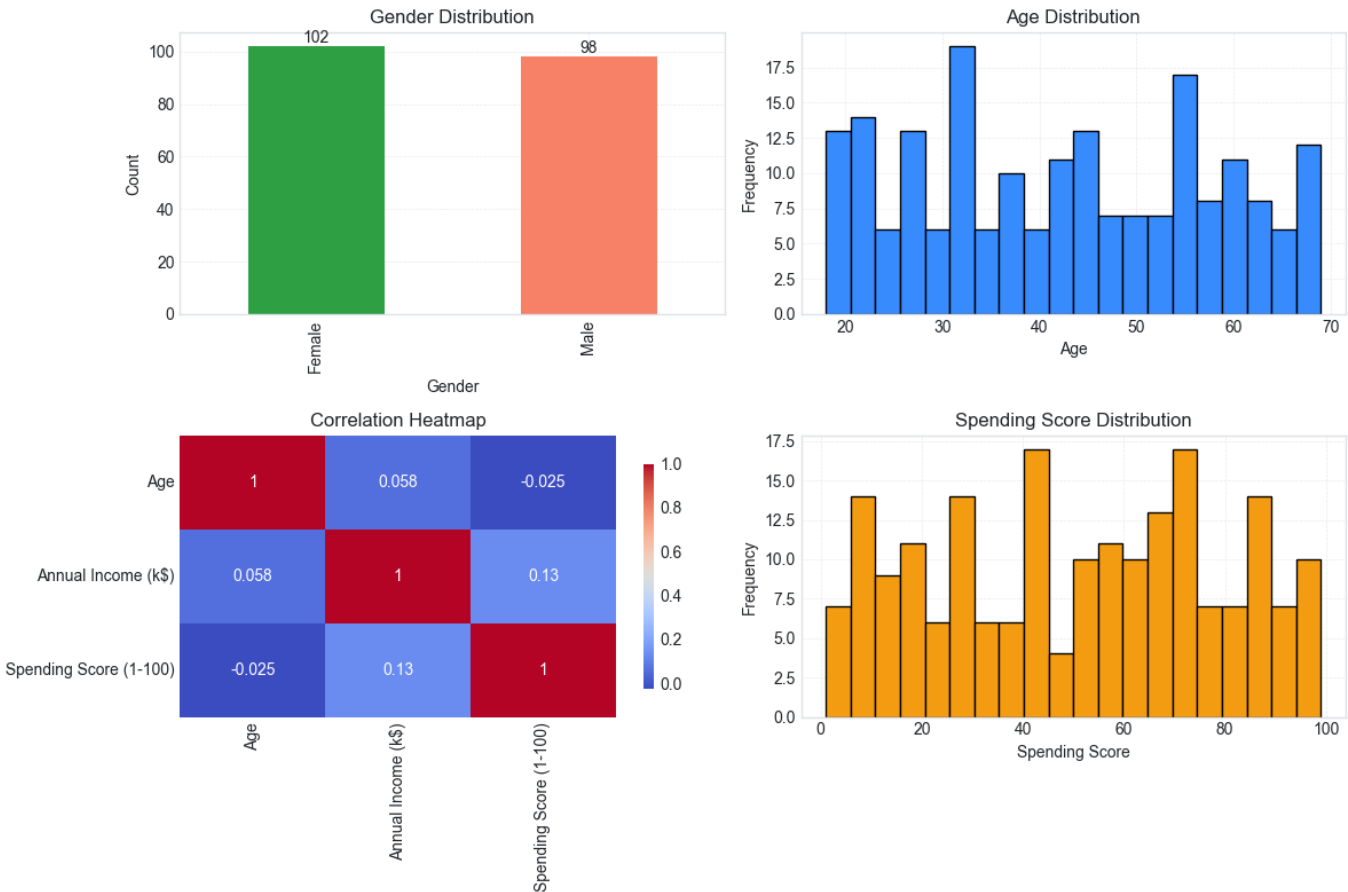


Lab 7: Customer Segmentation using Clustering Techniques

Dataset: Customer Segmentation Dataset (Mall Customers). Unsupervised learning task: group customers based on age, annual income, and spending behavior.

Exploratory Data Analysis

Total samples: 200
Features used: Age, Annual Income, Spending Score
Gender distribution: {'Female': 102, 'Male': 98}



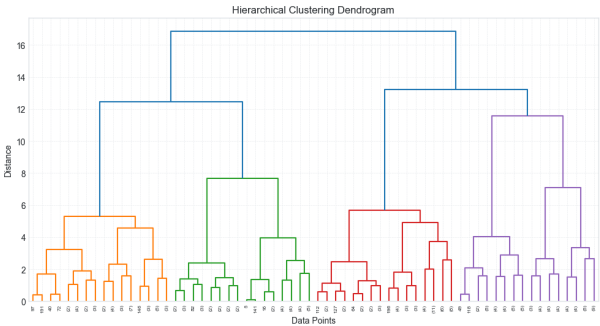
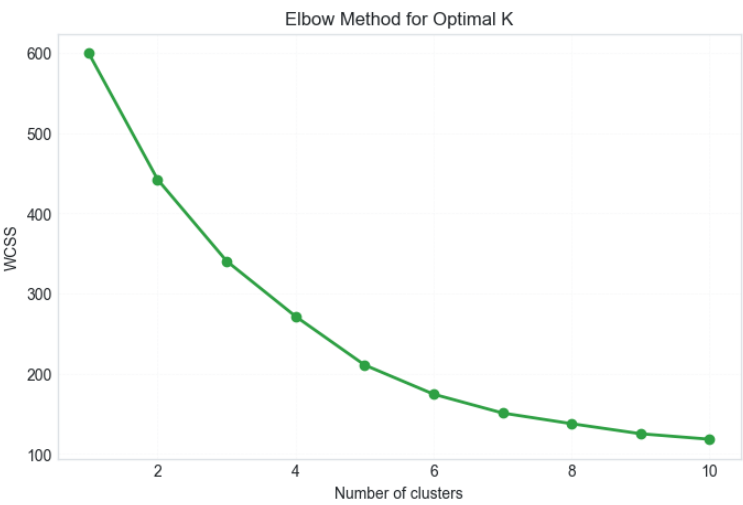
Clustering Implementation

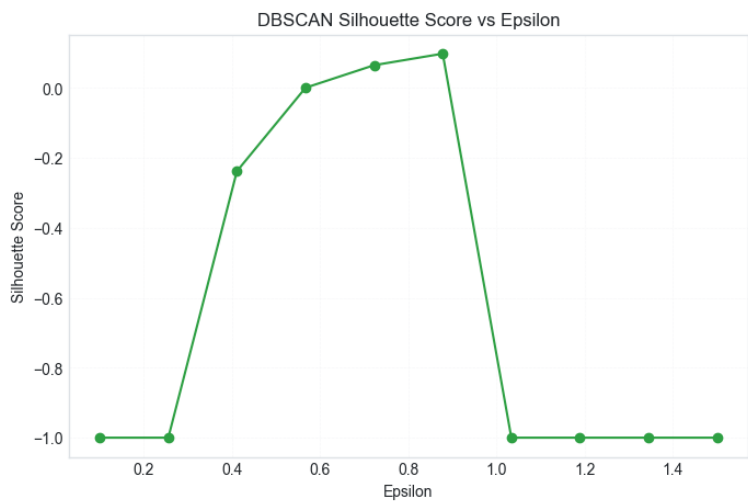
Algorithms Implemented: K-Means, Hierarchical (Agglomerative), DBSCAN

Optimal number of clusters (K-Means & Hierarchical): 5

DBSCAN Parameters: $\text{eps}=0.88$, $\text{min_samples}=5$

Noise points detected by DBSCAN: 1





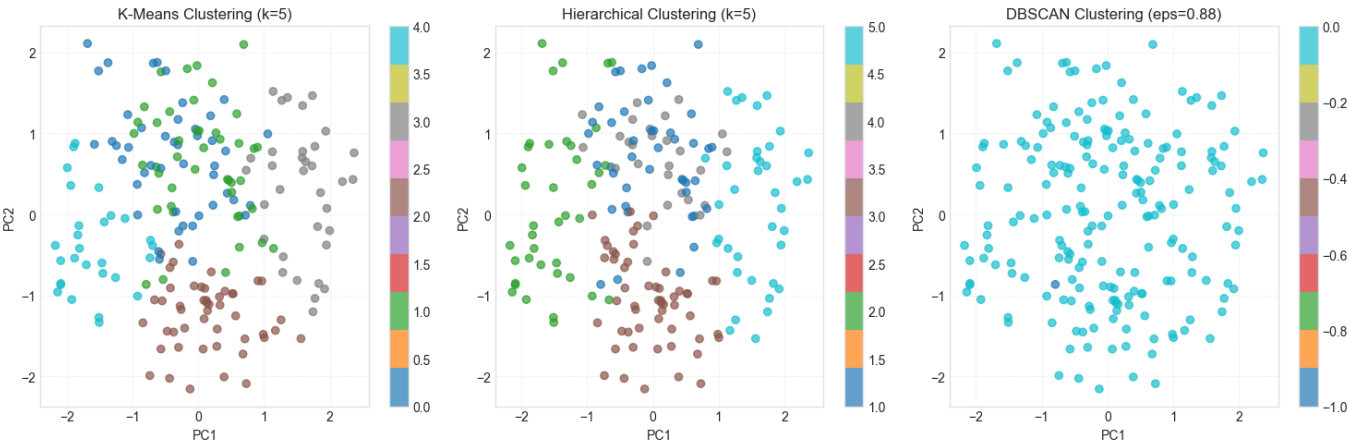
Performance Evaluation

Performance Metrics:

Clustering Performance Metrics

Algorithm	Silhouette Score	Davies-Bouldin Index	Calinski-Harabasz Index
K-Means	0.3158	0.9422	89.9687
Hierarchical	0.2914	1.034	80.4977
DBSCAN	-1.0	-1.0	-1.0

Cluster Visualization (PCA Projection)



Generated Images

- eda_overview.png
- elbow_method.png
- dendrogram.png
- dbscan_eps_tuning.png
- cluster_profiles.png
- metrics_table.png
- clustering_comparison.png